

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	BHUVANESHWARAN J	Reg. No.:	192511402
Section / Batch:	SLOT B	Date:	22/07/2026

Code of Conduct

I BHUVANESHWARAN J(192511402) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: BHUVANESHWARAN J

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == ']')) {
pop();
} else {
return 0;
}

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (isEmpty()) {  
2     return 0;  
3 }  
4 else if ((ch == '[' && stack[top] == '[') ||  
5         (ch == ']' && stack[top] == ']')) {  
6     pop();  
7 } else {  
8     return 0;  
9 }  
10 }
```


Test Input
stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q6 Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head  
2 while temp != NULL:  
3     print(temp.data)  
4     temp = temp.next
```


Test Input
stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUV

1925114

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q7 Stack

Point out the error.

```
POP(stack):
if top == 0:
    print("Underflow")
else:
    return stack[top--]
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack):
2     global top # Modify the global top variable
3     if top == 0:
4         print("Underflow")
5         return None
6     else:
7         top -= 1 # Decrement top first to get the correct index
8         return stack[top]
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 prev = NULL
2
3 while current != NULL:
4     nxt = current.next
5     current.next = prev
6     prev = current
7     current = nxt
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q9 Linked List

5 marks

What is the error in the below code?
new.next = head
head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 newNode.next=head
2 head=newNode
```

Test Input

stdin input (optional)

Output

Pending manual review

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

QUESTIONS

Q1

Binary Search

5 marks

Q2

Stack

5 marks

Q3

Stack

5 marks

Q4

Stack

5 marks

Q5

Stack

5 marks

Q6

Linked List

5 marks

Q7

Stack

5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):
2     if front == -1:
3         print("Underflow")
4         return None
5     else:
6         item = Q[front]
7
8         if front == rear:
9             front = -1
10            rear = -1
11 else:
12     front = front + 1
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q1 Stack Notations - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
    if (isdigit(expression[i])) push(expression[i] - '0');
    else {
        int op1 = pop();
        int op2 = pop();
        // perform operation: result = op1 + op2;
        push(result);
    }
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (expression[i] != '\0') {
2     if (isdigit(expression[i])) push(expression[i] - '0');
3     else {
4         int op2 = pop();
5         int op1 = pop();
6         // Perform operation: result=op2*op1;
7         push(result);
8     }
9 }
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1) {
3         return -1;
4     }
5
6     int data = queue[front];
7
8     if (front == rear) {
9         front = -1;
10    }
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {
2     if (l > r) {
3         return -1;
4     }
5     int mid = l + (r - l) / 2;
6     if (arr[mid] == x) {
7         return mid;
8     }
9     if (arr[mid] > x) {
10        return binarySearch(arr, l, mid - 1, x);
11    }
12    return binarySearch(arr, mid + 1, r, x);
13 }
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

192511402RB

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {
2     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
3         printf("Deque Overflow\n");
4         return;
5     }
6     if (front == -1) {
7         front = 0;
8         rear = 0;
9     }
10
11    else if (front == 0) {
12        front = MAX - 1;
13    }
14    deque[front] = data;
15 }
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

19251140288

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {
2     if (rear == MAX - 1) {
3         printf("Queue Overflow\n");
4         return;
5     }
6     if (front == -1) {
7         front = 0;
8     }
9     rear++;
10    queue[rear] = data;
11 }
```

Test Input

stdin input (optional)

Output

SIMATS

ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J

19251140288

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

QUESTIONS

Q1

Stack Notations - Postfix Expression

5 marks

Q2

Circular Queue

5 marks

Q3

Binary Search

5 marks

Q4

Queue

5 marks

Q5

Queue

5 marks

Q6

Doubly Linked List

5 marks

Q7

Stack

5 marks

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {
2     if (rear == MAX - 1) {
3         printf("Queue overflow\n");
4         return;
5     }
6     if (front == -1) {
7         front = 0;
8     }
9     rear++;
10    queue[rear] = data;
```

Test Input

stdin input (optional)

Output

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

Ask Gemini

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J 192511402RB

C CODE OUTPUT PREDICTION

QUESTIONS

- Q1 C Code - Pointer with Array 10 marks
- Q2 C Code - Linked List Node 10 marks
- Q3 C Code - Binary Search Complexity 10 marks
- Q4 C Code - Pointer Increment 10 marks
- Q5 C Code - Array Address 10 marks
- Q6 C Code - Linked List Traversal 10 marks
- Q7 C Code - Recursion (Factorial) 10 marks

Q2 C Code - Linked List Node 10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main()
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

20

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

Ask Gemini

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

BHUVANESHWARAN J 192511402RB

C CODE OUTPUT PREDICTION

QUESTIONS

- Q1 C Code - Pointer with Array 10 marks
- Q2 C Code - Linked List Node 10 marks
- Q3 C Code - Binary Search Complexity 10 marks
- Q4 C Code - Pointer Increment 10 marks
- Q5 C Code - Array Address 10 marks
- Q6 C Code - Linked List Traversal 10 marks
- Q7 C Code - Recursion (Factorial) 10 marks

Q3 C Code - Binary Search Complexity 10 marks

```
while(low <= high){
mid=(low + high)/2;
}
```

Your Answer

0(log n)

Save Answer